CSE558 · DATA SCIENCE — LECTURE 3

Describing & Cleaning Data

The mean and median tell you where. Today's tools tell you how spread out, how lopsided, and how messy.

TODAY'S ARC

Spread → **Shape** → **Plots** → **Cleaning**

RECAP

Where we left off

1

Four attribute types

Nominal, ordinal, discrete, continuous.

2

Central tendency

Mean (sensitive to outliers), median (robust), mode.

DISCUSSION — 5 MINUTES

The dilemma

You run a typical IT company and want your report to show that a typical worker is paid well.

SCENARIO 1

A handful of employees earn very low salaries. As CEO, do you report the mean or the median salary? Why?

SCENARIO 2

Now suppose instead a lot more employees earn very low salaries, not just a handful. Does your answer change? Why?

Choosing a statistic is never neutral. The same dataset can support two honest-sounding but very different stories.

There's no single 'correct' answer here — the goal is to making most appropriate choice.

LAST DAY QUIZ

Check your understanding

Q1 A dataset of 1000 salaries has one billionaire in it. The best measure of the 'typical' salary is:

Median

Q2 Which attribute type can you meaningfully average?

Continuous (& sometimes discrete)

Q3 One-hot encoding is the natural fit for which attribute type?

Discrete

Now

Today

1

Four attribute types

Nominal, ordinal, discrete, continuous.

2

Central tendency

Mean (sensitive to outliers), median (robust), mode.

3

Today's question

The mean told us "where." It said nothing about how they are spread out.



BEFORE THE FORMULAS — A WARM-UP

Same mean, completely different story

Team A — daily commits

48, 49, 50, 51, 52

Everyone commits reliably, every day.

Mean = 50

Team B — daily commits

10, 30, 50, 70, 90

Wildly inconsistent — some days nothing, some days a flood.

Mean = 50

Both teams have the exact same mean. But the “spread” is very different.

DISPERSION

Variance & standard deviation

Let $x_1, x_2, \ldots, x_N$ be N values. The variance is:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

Standard Deviation: $\sigma = \sqrt{\sigma^2}$.

BACK TO OUR TWO TEAMS

Team A (48–52): tiny variance.

Team B (10–90): huge variance.

Why square the distances?

Distances above and below the mean would otherwise cancel out to zero. Squaring makes everything positive first.

TRY IT YOURSELF

Compute variance

THE DATA — SAME TEAM AS LECTURE 2

Code-review turnaround times (hours): 3, 5, 4, 22, 6 — mean = 8

Variance = 50

Std dev ≈ 7.07 hours

A standard deviation nearly as big as the mean itself.

Mean absolute deviation

A simpler alternative to variance — skip the squaring, just average the absolute distance from the mean:

$$\mathbf{MAD} = \frac{1}{N}\sum_{i=1}^{N}|x_i - \bar{x}|$$

QUICK COMPARISON

Our code-review 3, 5, 4, 22, 6:

MAD = (5+3+4+14+2)/5 = 5.6

Std dev ≈ 7.07

Std dev is larger than MAD.

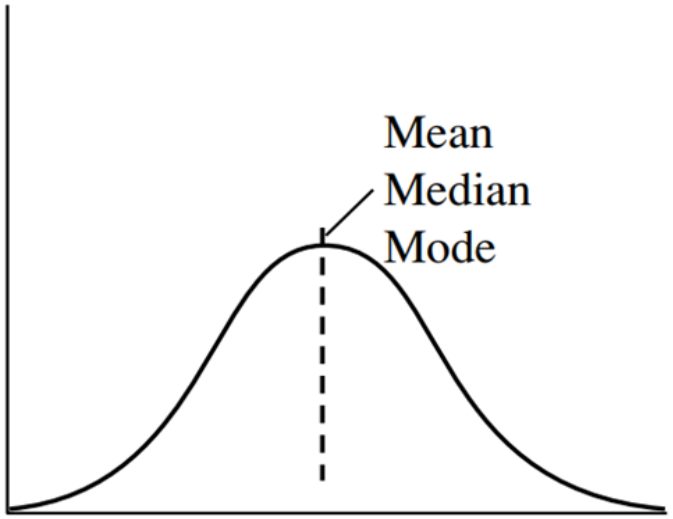
Is std dev always greater than MAD?

Std dev is popular. But why?

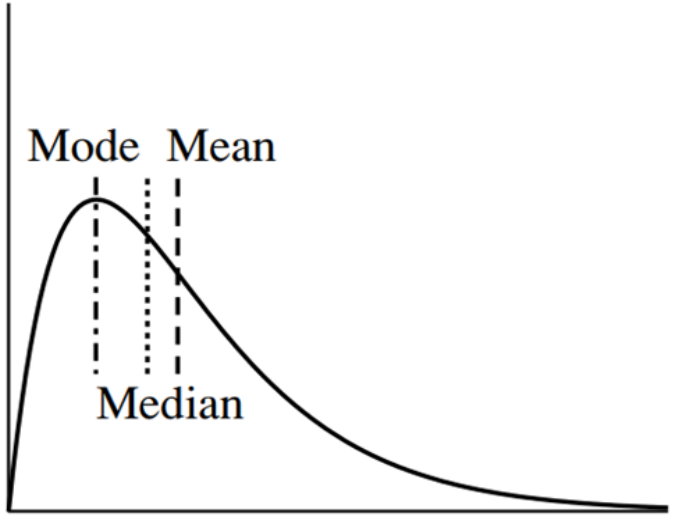
Squaring is differentiable everywhere, which matters a lot in optimizing models.

Skewness

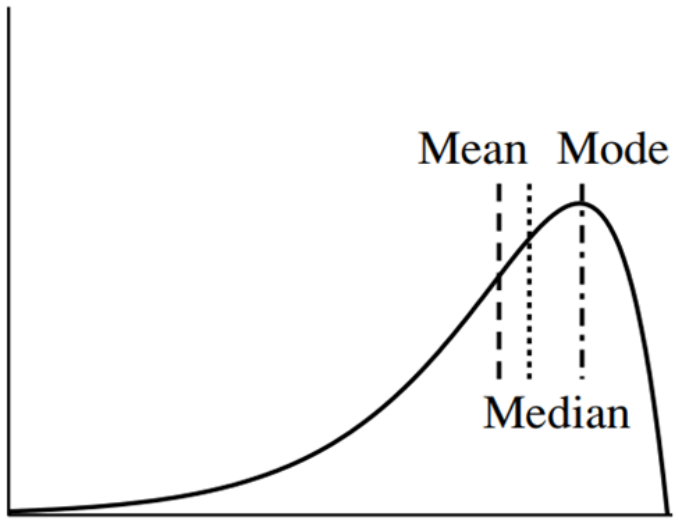
Measures the nature of a distribution's symmetry — and notice where mean, median, and mode fall relative to each other.



Symmetric



Positively Skewed



Negatively Skewed

Positively skewed: long tail to the right. Mode < Median < Mean — the big values on the right pull the mean up.

Negatively skewed: long tail to the left. Mean < Median < Mode — the reverse.

DISCUSSION — 2 MINUTES

Name a skewed dataset

KARL-PEARSON COEFFICIENT

A number that quantifies skew, not just its direction:

Coefficient < 0 → negatively skewed

Coefficient = 0 → symmetric

Coefficient > 0 → positively skewed

QUESTION

Give one example each of positively and negatively skewed data and say why.

$$kp_1 = \frac{\bar{x} - \text{Mode}}{\sigma}$$ within (-1,+1)

$$kp_2 = \frac{3(\bar{x} - \text{Median})}{\sigma}$$ within (-3,+3)

Correlation

Does one variable depend on the other?

Before quantifying correlation,

1. Remove outliers, if any.
2. Check that the distributions are close to normal.

Recall from Lecture 1

“If it rains, there's more traffic” was our first example of correlation.

Where this shows up in your project

If two columns in your dataset move together, that's a hypothesis worth investigating — not a conclusion you can report as-is.

QUICK POLL

Spot the fake correlation

For each pair, vote: does A cause B, does B cause A, or is a hidden third factor driving both?

Ice cream sales ↑ ⟷ **Electricity Bills ↑**

Hidden factor

Hot weather drives both — nobody buys ice cream and it increases the electricity bill.

More firefighters at a fire ⟷ **More damage caused**

Hidden factor

Bigger fires cause both — more firefighters are sent AND more damage happens.

Hours studied ⟷ **Exam score**

A likely causes B

Still not proven by correlation alone — but a controlled study could confirm it.

Range & quantiles

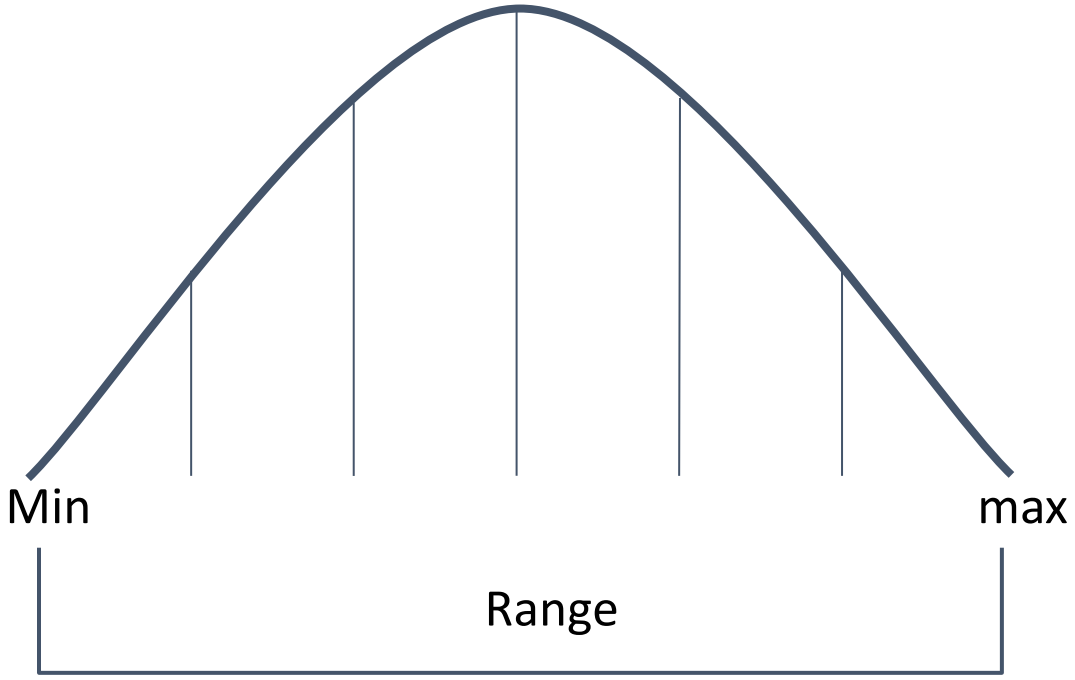
RANGE

Range = max − min

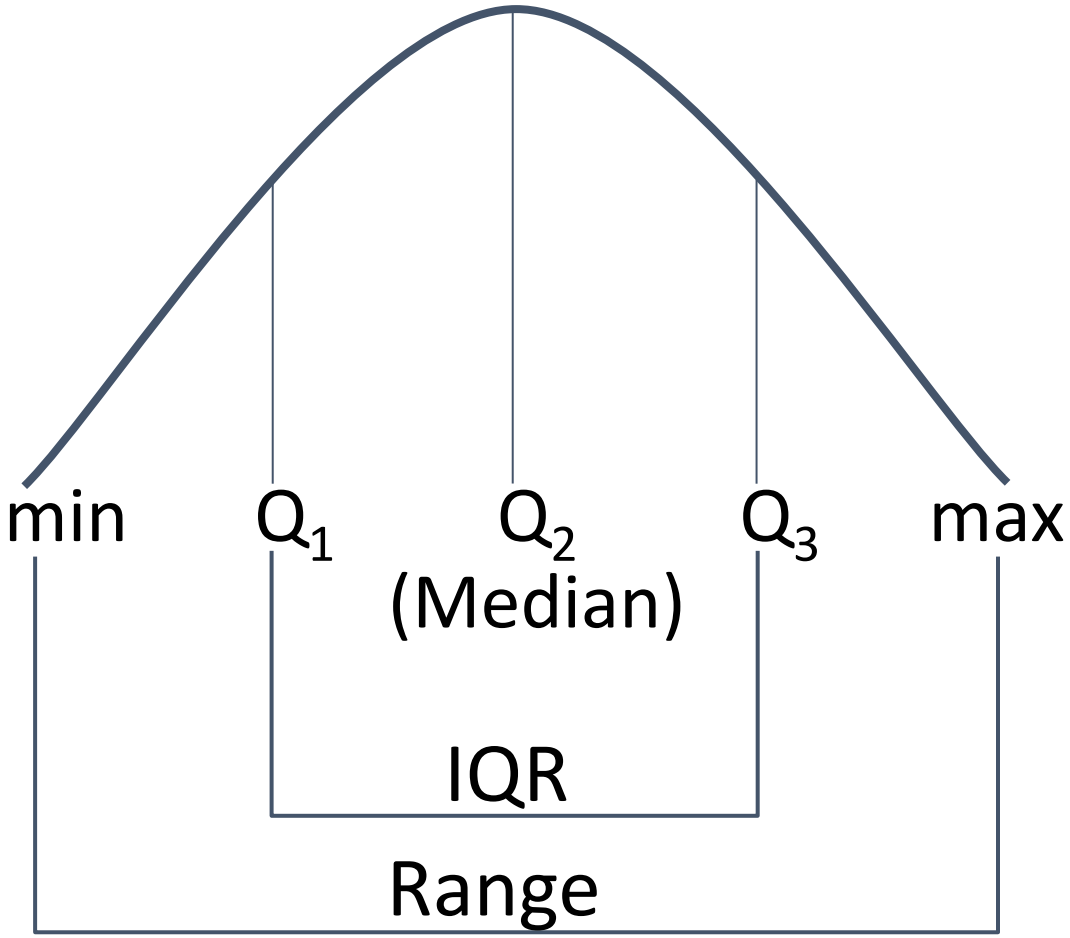
Example: values 1, 3, 4, 2, 7, 6, 5 → max=7, min=1 → Range = 6.

6-QUANTILE

Split sorted data into x equal-sized groups instead of just min/max — a finer-grained view of the shape.



Quartile



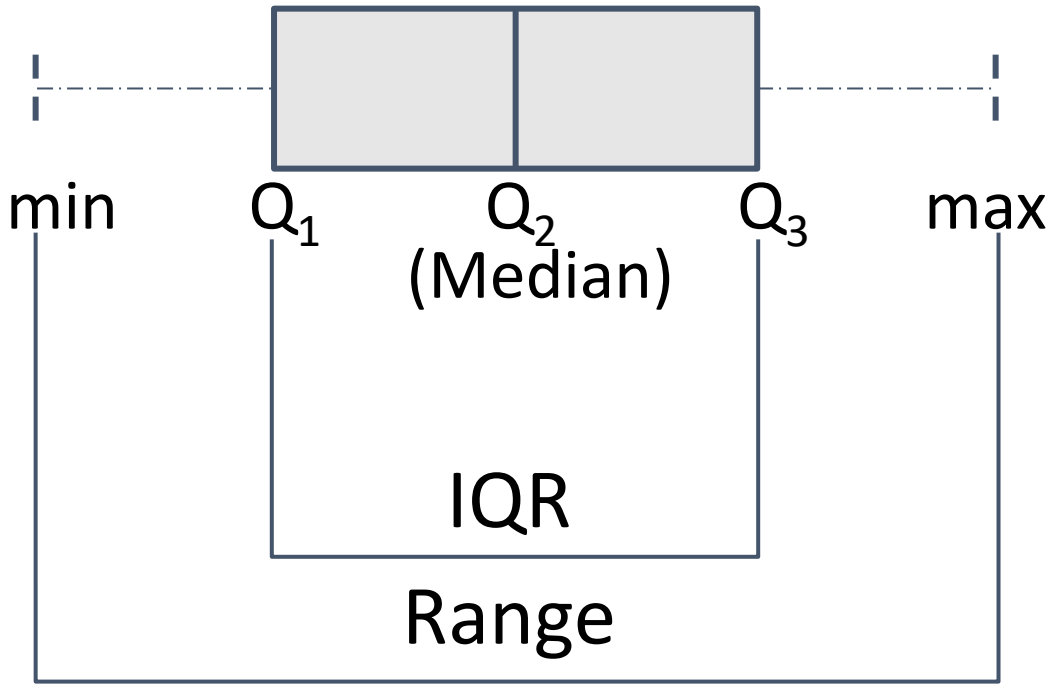
THE FIVE-NUMBER SUMMARY

The version everyone actually uses — four quarters instead of x:

- **Minimum**
- **Q1 — 25th percentile**
- **Q2 — median (50th percentile)**
- **Q3 — 75th percentile**
- **Maximum**

IQR = Q3 − Q1 — the spread of the "middle half" of your data, unbothered by extreme outliers.

Quartile



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Build a boxplot, live

Twelve exam scores: 25, 80, 35, 30, 60, 20, 65, 70, 45, 85, 90, 75. Sort them, then find the five landmarks yourself before checking the illustration below.

Sorted: 20, 25, 30, 35, 45, 60, 65, 70, 75, 80, 85, 90. Range = 90−20 = 70. IQR = 77.5−32.5 = 45.



What a boxplot actually tells you

Distribution, at a glance

The box shows where the middle 50% of your data lives. A compact and simple graphic to realize the shape.

Skewness

If the median line sits off-center in the box, or one whisker is much longer, the data is skewed in that direction.

Outliers

Points far beyond the whiskers (typically past 1.5× IQR from the box) are flagged as candidate outliers.

A CAUTIONARY TALE

When boxplots mislead

THE BLIND SPOT

Bimodal data

Imagine exam scores where half the class scored around 40 and the other half around 90 — two distinct clusters, nobody in between.

The five-number summary only describes position — it says nothing about shape between the landmarks.

THE FIX

Pair it with a histogram

A histogram of the same scores would immediately show two separate peaks. The bimodality a boxplot can never reveal on its own.

The rule of thumb

No single plot tells the whole story. A boxplot for outliers and quartiles, a histogram for shape — use them together, not instead of each other.

ONE MORE LAYER OF SHAPE

Moments & kurtosis

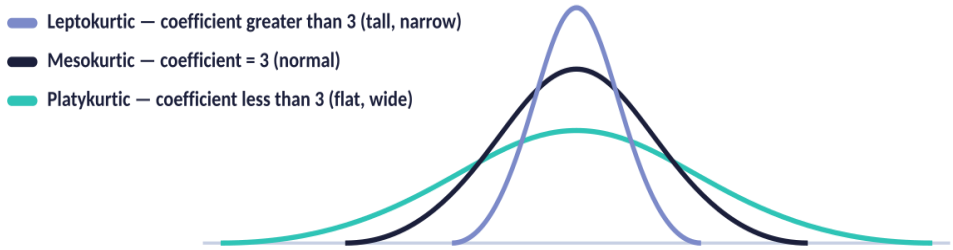
$$\frac{1}{N}\sum_{i=1}^{N}(x_i - 0)^r \qquad \mu_r = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^r$$

MOMENTS

Every statistic you've learned today is secretly the same family, counted higher and higher:

- 1st around origin = the mean, ie $\bar{x}$
- 2nd around mean = the variance, ie σ^2
- 3rd around mean $\approx$ skewness, ie $\frac{\mu_3}{\sigma^3}$
- 4th around mean $\approx$ kurtosis, ie $\frac{\mu_4}{\sigma^4}$

KURTOSIS — measures the flatness of a distribution



KURTOSIS COEFFICIENT

Leptokurtic — coefficient > 3 — tall, narrow, heavy-tailed

Mesokurtic — coefficient = 3 — a typical normal distribution

Platykurtic — coefficient < 3 — flat, light-tailed

Scatter plots

Plots 2- or 3-dimensional data — one point per object.

Understand correlation

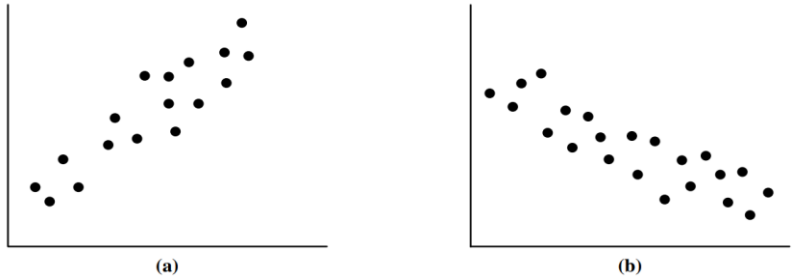
Positive, negative, linear, non-linear — the shape of the cloud tells the story.

Spot patterns

Clusters and outliers jump out visually in a way a table never shows.

Watch for overplotting

Thousands of points stacked on top of each other hide the real density is a problem.



SAME WARNING, ONE MORE TIME

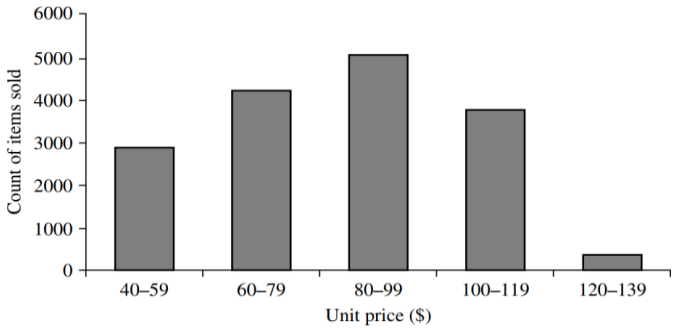
A scatter plot showing a clear trend is still not proof of causation.
It's the single most persuasive-looking chart in statistics and the easiest to over-trust!!

Categorical & numeric plots

CATEGORICAL — ABSOLUTE

Bar graph

Height shows the raw count in each category. Best when actual quantities are important.



CATEGORICAL — RELATIVE

Pie chart

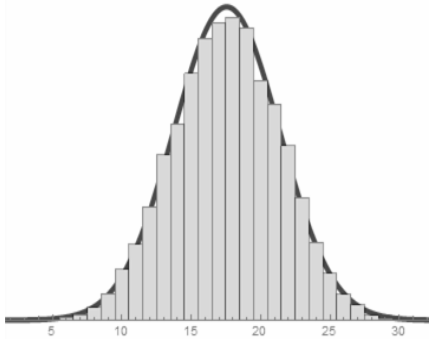
Slice size shows the share of the whole. Best when the story is “what fraction,” not “how many.”



NUMERIC — DISTRIBUTION

Histogram

Bins a continuous range and counts frequency in each bin — your first real look at a distribution's shape.



Comparing distributions

QUANTILE PLOT

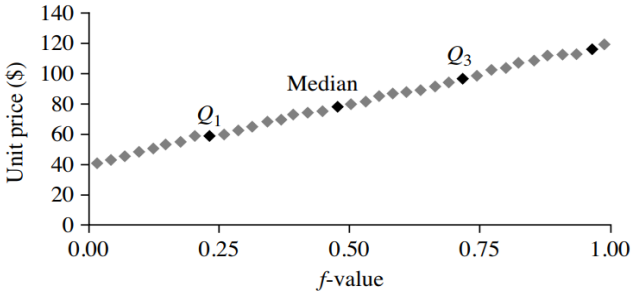
Visualizes a single distribution's shape, and lets you compare it against a reference (like a normal distribution).

HOW IT'S BUILT

1. Sort the data.
2. Compute each point's percentile (f-score).
3. Plot value against percentile.

$$f_i = \frac{i - 0.5}{N}$$

Unit price ($)	*Count of items sold*
40	275
43	300
47	250
—	—
74	360
75	515
78	540
—	—
115	320
117	270
120	350



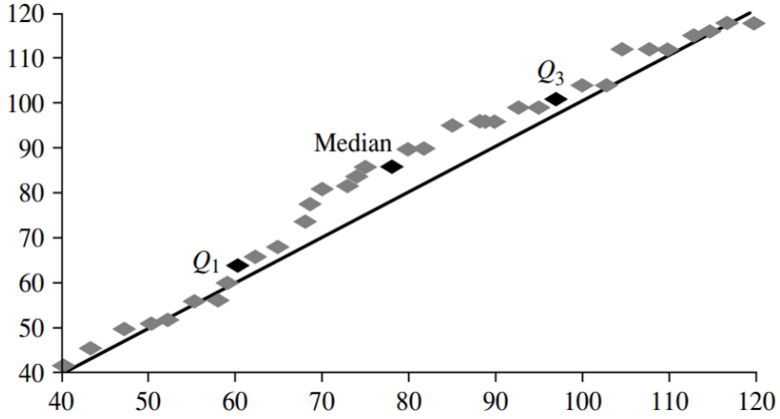
Comparing distributions

Q-Q PLOT

The natural next step: instead of comparing one distribution to a theoretical reference, compare two real distributions directly against each other.

If both distributions are identical in shape, the points fall on a straight diagonal line. Any curve or bend is where they differ.

Example use: are this year's exam scores distributed the same way as last year's?



Handling outliers

1 **Sort the data**

The simplest way to spot extreme values at either end.

2 **Visualize it**

Boxplots, quantile plots, scatter plots, histograms — every tool from today's plot gallery applies here.

3 **Project to fewer dimensions**

For high-dimensional data, techniques like PCA can reveal outliers invisible in any single column.

4 **Normalize**

Rescale so no single attribute's raw magnitude dominates the picture.

Domain knowledge is always an added advantage — and treat this as iterative.
After each fix, ask again: does an outlier still exist?

Handling missing values

First, understand why values are missing — randomly, or systematically (which is far more dangerous and easy to miss).

OPTION 1 — DELETION

Drop the row (or column) entirely. Simple, safe if missingness is rare and random — wasteful if it isn't.

The trap: if missingness is systematic (e.g. wealthier respondents skip an income question), deleting those rows quietly biases everything downstream.

OPTION 2 — IMPUTATION

Fill it in with an educated guess:

a. Central tendency — the column's mean, median, or mode

b. Average kNN — the average of similar rows' values

c. Regression — predict the missing value from other columns

Handling noise

Data = Signal + Noise

Noise is the random, meaningless variation riding on top of the real pattern you actually care about.

Binning

Group nearby values into buckets and smooth within each bucket — trades precision for stability.

Regression

Fit a smooth curve or line through the data; treat the residual as the noise you're removing.

Dimension reduction

Some noise only appears when you remove the low-signal directions in high-dimensional data.



TRY IT YOURSELF — 3 MINUTES

Data cleaning triage

A SURVEY DATASET'S “AGE” COLUMN

For each row, decide: clean, outlier, missing, or noise — and what you'd actually do about it.

- **24**
- **999**
- **(blank)**
- **“twenty-six”**
- **31**

REVEAL

24 → Clean

999 → Outlier

(blank) → Missing

“twenty-six” → Noise

31 → Clean

The “noise” row isn't wrong data — it's a formatting inconsistency. Standardize it to 26 before it silently breaks anything downstream.



STUDY MATERIAL

Reference

Data Mining: Chapter 2.2 by Han, Kamber & Pei

Slides